

Corporate Governance Reform and Dividend Policy: Does the Severity of Penalties Matter?

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Abstract

Do mandatory provisions of corporate governance reform (CGR) replace payout policy used as a signal of better corporate governance in emerging market? We answer this question by exploiting two different exogenous shocks of the same listing regulation in the emerging market context of India. The first shock relates to the adoption of CGR with reputational penalties and the second to the additional imposition of personal liability of corporate insiders for non-compliance. Our results show that firms that are affected by CGR reduce their dividend payouts, on average, by at least 3%. However, this governance replacement effect is not observed following the initial adoption but only after the imposition of additional personal liability in the form of stricter financial and criminal penalties. Our findings highlight the merit of increasing personal penalties in bringing adequate deterrence of non-compliance that substitutes of dividend payouts as a costlier corporate governance signal in an emerging market.

Keywords: Dividend payout; Corporate governance reform; Stricter penalties; Natural Experiment

1. Introduction

Agency and information related problems have been the major drivers of corporate dividend policy in both emerging and developed economies (Easterbrook, 1984; La Porta et al., 2000; John et al., 2015 among others). ‘Substitution argument’ of corporate payout policy establishes corporate governance reforms (CGRs) and higher payout policy as substitutes for reducing agency (and information) costs of free cash flows (John et al., 2015); Caton et al., 2016)¹. Firms in emerging markets, unlike their developed market counterparts, face relatively severe agency and information related problem between corporate insiders and minority outsiders with the existence of weaker market forces of corporate scrutiny (Claessens and Yurtoglu, 2013). Similarly, in these markets, better governed firms use higher Dividend Payout Ratios (DPRs) as a substitute to weaker external corporate governance (Pinkowitz et al., 2006).² Therefore, the signalling property of DPR should become less relevant following exogenous improvement in CGR in an emerging market.

As emerging markets seek to improve their capital markets and attract foreign investors, they often adopt CGR provisions similar to those in developed markets (Martynova and Renneboog, 2011). However, despite the importation of the well-established regulatory framework from developed markets, there are concerns on the effectiveness of enforcement to deter non-compliance and improve internal corporate governance practices. In an extensive survey of the literature, Claessens and Yurtoglu (2013) note that on average, the effectiveness

¹ Even though share-repurchases have dominated the corporate sphere of developed markets over cash dividends in the recent years owing lower pre-commitment cost and tax related advantage (Caton et al., 2016), cash dividends are still the most popular payout method in emerging markets (Bakaert and Harvey, 2000; Jeon et al., 2011).

² Similarly, a survey of CFOs of Indian firms by Anand (2004) shows that dividend payout act as signalling mechanism of firm’s performance, prospects and corporate governance. The survey was administered in the year 2001 and is comparable with our study period.

of CGR enforcement in emerging economies is substantially lower in comparison to their developed market counterparts. Similarly, Dharmapala and Khanna (2013) note that the effectiveness of CGR enforcement in an evolving regime of emerging market is affected by the adequacy of CGR sanctions and associated personal liabilities of the insiders/managers in the events of non-compliance. Therefore, when CGR is introduced with inadequate penalties, the adoption of CGR alone might not stimulate changes in corporate behaviour and therefore CGR may not substitute higher DPR as governance tool.

To the extent that CGR does substitute higher payout policy as governance tool, in this study, we investigate whether it is the CGR that imposes reputational penalties or the accompanying harsher sanctions that increase the personal liability and criminal penalties of corporate insiders that replace DPR as governance signal in emerging markets. Our distinction based on the severity of the penalties of the enforced CGR allows us to examine relative merits of harsher personal punishments vis-à-vis reputational penalties on CGR's non-compliance in a relatively weaker and evolving regulatory environment.

Economic reasoning suggests that adoption of mandatory CGR that introduces reputational penalties should exhibit substitutive effect on corporate DPR (Jensen et al., 1992; Noronha et al., 1996; Borokhovich et al., 2005; John et al 2015) as these reputational penalties are large enough to induce higher compliance. However, if CGR is not widely adopted by firms or seen as credible by external stakeholders to adequately induce more complying behaviour among firms there could still be a role for DPR as a signal of corporate governance quality. It is only when CGR imposes sufficiently deterring penalties that CGR translates into meaningful improvement in corporate scrutiny and therefore should have an impact on DPR. We test this prediction using two corporate governance regulatory shocks in the Indian market.

Prior to year 2000 the Indian corporate governance framework was largely informal (Dharmapala and Khanna, 2013). In the absence of mandatory CGR controlling shareholders can find it challenging to convince its stakeholders about their intention of adopting good governance practices. Therefore, Indian firms would pay higher DPR to signal the better quality of their internal governance given the weaker external governance environment to external stakeholders.

In 2000, India introduced CGR of greater disclosure and board independence in the stock market listing agreement, popularly known as Clause-49. Clause-49 was intended to reduce agency problems and introduce sound corporate governance practices. As such, following the ‘substitution argument, we would expect that the firms that were obliged to comply, relative to those who are not, should rely less on DPR as corporate governance signal. The initial penalty for non-compliance was delisting from the stock market, which constitutes significant reputational penalties but without any personal liability of corporate insiders with financial and criminal penalties. However, in the year 2004 the regulators further amended section 23E to impose severe financial and criminal penalties for violating regulatory provisions of Clause-49. Following Becker’s (1968) argument that harsher punishment can lead to effective compliance behaviour in an evolving regulatory environment, amendment of section 23E allows us to investigate whether additional imposition of personal liability with financial penalties and criminal charges can have a greater impact on DPR relative to the initial reputational penalty of CGR in the year 2000.

Exploiting both the Clause-49 introduction and section 23E amendment, which exogenously generate affected (hereafter treated) and unaffected (hereafter control) firms, and using a sample of listed firms spanning 1997-2007, the difference-in-difference (DiD) estimations report the following two main findings. First, the introduction of Clause-49 in

2000 seems to have no impact on the DPR of treated firms. However, the Section 23E amendment leads to a significant reduction in DPR. In quantitative term, the treated firms have reduced their DPR, on average, by 3% to 5%, depending on specifications. These results are robust to several additional checks. Our results support dividend substitution hypothesis of CGR as the treated firms replace DPR as governance tools after Clause-49, but only after the imposition of greater personal liability of insiders in the form of severe financial and criminal penalties in Section 23E that would deter non-compliance, consistent with Becker's (1968) punishment model.

Second, examining the cross-sectional heterogeneity our results show that within treated firms, the DPR of high dividend paying firms have reduced significantly in comparison to low dividend paying firms. The impact of Clause-49 replacing DPR as signal for high levels of corporate governance is more applicable to firms which initially pay higher dividends. This result provides some support to the view that firms use high DPR to convey their governance quality in the absence of effective country-level CGR. Similarly, after the imposition of 23E, highly-levered treated firms also reduce their DPR by a greater degree than those treated firms with lower leverage. Therefore, this suggests that the stricter penalties of Section 23E benefit creditors indirectly as managers can use the savings generated from the lower DPR to service their company's debt.

Our study contributes to the following strands of literature. First, to the best of our knowledge, this is the first study to link and test the dividend 'substitution argument' and Becker's (1968) punishment-governance model in a unique single institutional setting. Specifically, we investigate whether it is the adoption of internationally established governance regulations or the imposition of harsher and economically large punishments for non-compliance of these regulations that helps firms to replace DPR as a governance tool.

Second, our study also adds to the literature on the legal determinants of DPR in emerging economies (Aivazian et al., 2003; Mitton, 2004). Aivazian et al. (2003) show that whereas firm-level determinants of DPR of emerging markets are similar to those of developed markets, country-level differences are an important determinant of DPR in these markets. In a cross-country study of emerging markets, Mitton (2004) finds a positive relation between corporate governance and DPR for a limited subsample of countries with strong external corporate governance regulations. In our study, we show that the CGR is effective in substituting DPR as governance tool in an emerging market context when CGR is accompanied with stricter monetary and criminal penalties.

The rest of the paper is organised as follows. Section 2 briefly describes the characteristics of Clause-49 and amendment section 23E. Section 3 covers prior literature on DPR and CGR and develops the testable hypotheses. Section 4 describes the dataset followed by the discussion of empirical results in section 5. Section 6 concludes.

2. CGR in India

2.1. Clause-49

The corporate governance environment in India was largely informal prior to the induction of Clause-49 in 2000 (Dharmapala and Khanna, 2013). As Indian companies began to seek external financing this led for the need for a sound corporate governance regulatory framework to ensure better investor protection. In 1998, the Confederation of Indian Industry introduced the voluntary Corporate Governance Code, which was adopted by only few major companies. Consequently, the Code evolved into the mandatory Clause-49 provisions in February 2000 applicable to wider spectrum of listed companies in India. Clause-49 of the stock exchange listing agreement is a set of corporate governance reforms enacted by the

Securities and Exchange Board of India (SEBI), the governing body of listed companies in India. Clause-49 introduced greater compliance, disclosure, board independence and transparency. The main goal is to improve the governance of publicly listed Indian firms by imposing greater disclosure requirements and enhancing the independence of boards and audit committees.

As shown in Figure 1, Clause-49 had a phased-in implementation schedule – firms with a listing flag “A” on the Bombay Stock Exchange (BSE) (i.e. the largest firms) had to comply by March 31, 2001 (Group 1 firms); companies with paid-up share capital of more than 100 million or net worth of more than 250 million at any point in time needed to comply by March 31, 2002 (Group 2 firms); companies with paid-up share capital of more than 30 million had to comply by March 31, 2003 (Group 3 firms). Additionally, firms that listed after 2000 for the first time had to comply with the reforms immediately, irrespective of their paid-up capital or net worth.

Figure 1 about here

The initial penalties for violating Clause-49 included reputational penalty of stock delisting from the stock exchange. However, in 2004 Section 23E was added to the Securities Contracts (Regulation) Act of 1956 which imposes more severe financial and criminal penalties for non-compliance of Clause-49.

2.2. Provisions of Clause-49 and DPR

We identify three major provisions in Clause-49 that could impact on DPR. These are: board-independence, independence of audit committees, and certification by CEO/CFO. Independent directors are valued by shareholders as they can prevent insiders from diverting

cash flows (Nguyen and Nielsen, 2010). Schellenger et al. (1989) propose that outside directors, with their monitoring capabilities, can be effective substitutes for DPR. The role of independent directors can be even more relevant in emerging markets characterized by higher information and agency related problems (Claessens and Yurtoglu, 2013). Clause-49 increases independent directors' marginal value by assigning them crucial roles, such as sitting on audit committees. The mandatory provision of board independence in Clause-49 may therefore reduce the role DPR as governance tool.

Similarly, Clause-49 requires an audit committee with the minimum of three directors, two-thirds of which are required to be independent and at-least one with experience in financial management. The clause also requires certification by the auditor or company's secretary on the compliance of corporate governance provisions and disclosures. An increase in independence of the audit committee reinsures investors of the accuracy of the reports and thus managers can afford to reduce dividends as they no longer need to use DPR as a governance mechanism (Schellenger *et al.*, 1989).

The third important provision in Clause-49 related to DPR, is that certifications on financial statements and internal control mechanism by the CEO/CFO are mandated, which increases personal accountability of insiders/managers in case of non-compliance. CEO/CFO certifications can therefore be argued to alleviate information and agency related problems of firms in emerging markets. Clause-49 can therefore lessen the need for DPR as governance signals. Taken together these provisions on the structure and accountability of board, the audit committee and management team certification increase firms' scrutiny from the external stakeholders and therefore could substitute DPR.

3. Empirical Hypotheses

In this section, we develop two testable hypotheses on the relation between CGR and DPR. The first relates to the effect of DPR on the adoption of Clause-49 in 2000 and second on the subsequent imposition of financial and criminal penalties in 2004.

3.1. *The Substitution Argument and the Relation between CGR and DPR*

Easterbrook's (1984) model argues that dividends act as a strong signal of financial performance of the firm and thus increases investor confidence leading to a stable flow of investments. Dividends can, therefore, substitute for missing legal protection of shareholders in emerging markets and firms pay higher dividends when they wish to establish a reputation for fair shareholder treatment (La Porta *et al.*, 2000). In countries with weak legal protection of minority investors, the reputational effect accounts for a significant part of the stock valuation of listed companies (Gomes, 2000). Managers can increase DPR to signal outside stakeholders of the company's current governance standards and its future financial potential (Easterbrook, 1984). In India, conveying strong corporate governance is arguably more important, due to the existence of weaker market based corporate scrutiny. In support of this argument, Smith and Watts (1992) report that when the strength of the signal increases with the information asymmetry between investors and managers, firms with larger information disparities should distribute higher dividends. However, when reforms on a national level are designed to reduce information disparities, and firms should no longer need to increase their dividends to the same extent as Clause-49 imposes disclosure requirements.

John *et al.* (2015) and Caton *et al.* (2016) postulate 'Substitution argument' in which sound corporate governance is a substitute for cash payout policies and vice versa. They argue that the scrutiny provided by higher payout essentially replaces internal corporate

governance mechanisms. This is consistent with Easterbrook (1984)'s conjecture that corporate governance tools and higher payout do indeed serve a common purpose of reducing manager-shareholder conflicts. Regular cash dividends can discipline managers by forcing them to generate enough cash so that the firm's payout policy can be met. As a result, managers would often have to access the external capital markets for additional funding, which in turn brings in outside investor scrutiny. However, John et al. (2015) notes that devising higher internal corporate governance may be costlier for firms, which specifically resonates with firms operating in emerging markets in the face of weaker market-forces of corporate scrutiny and evolving regulatory regime. Therefore, well governed firms in emerging environment in general pay higher dividends as a substitute to missing external corporate governance (Pinkowitz, 2006).

The introduction of CGR that improves external corporate governance should replace DPR as governance tool as dividend has cost of pre-commitment and tax related costs (Caton et al., 2016). In this new regime of better external governance in an emerging market, we maintain that CGR substitutes DPR and provides companies with financial flexibility as they do not have to pre-commit cash dividends to shareholders.

Thus, Clause-49 is expected to substitute DPR as a corporate governance mechanism since managers no longer need to pay higher DPR to reassure investors that they will not be expropriated (Schellenger *et al.*, 1989). Since the corporate governance environment in India was informal prior to the induction of Clause-49, this intervention should induce significant dividend reductions in the treated firms by the clause relative to the control firms and therefore we propose to our first hypothesis:

H₁: Following the adoption of Clause-49 in 2000, treated firms should reduce their DPR more than the control firms.

3.2. Harsher Sanctions and DPR

The role of corporate governance enforcement is well documented in the literature. Better enforcement increases managers' commitment power to investors and as a result, their firms are able to obtain low-cost external financing (Berglof and Claessens, 2004; Dutcher 2005). Potential shareholders invest more in firms when their rights are better protected (La Porta et al., 1999; Jackson and Roe, 2009). Even though a reputation for fair treatment of investors can help firms obtain external financing, law and enforcement are the ultimate differentiating factors that help explain why managers raise more funds in some markets than in others (La Porta et al., 1999). One of the ways of improving effectiveness of enforcement is to provide significant deterrence of non-compliance of CGR through the imposition of harsher sanctions (Dutcher, 2005; Wang et al., 2017).

The evidence on the relative merit of harsher sanctions and personal penalties over reputation penalties to improve effectiveness of corporate governance interventions is mixed. One strand of literature argues that reputational penalties are economically very large and therefore should be an adequate deterrence when compared to legal and private penalties (Karpoff and Lott, 1993).³ If this argument holds in our case, the imposition of stringent financial and criminal sanctions in 2004 should be redundant and the adoption of CGR in 2000 that introduced reputational penalties of stock-delisting should provide adequate punishment to ensure compliance, supporting hypothesis 1.

However, there are other empirical studies that support the use of stricter sanctions to

³ Gneezy and Rustichini (2000) finds that monetary fines can increase non-compliance, contrary to the intended higher compliance.

induce higher compliance of new regulations. For example, Dutcher (2005) argues that only sanctions that introduce substantial criminal and financial penalties can adequately deter corporate non-compliance. Similarly, Wang et al. (2017) show that in emerging market context of China, the stock market perceives punishments with monetary penalties as more severe than those without monetary penalties. Dharmapala and Khanna (2013) show that stricter financial and criminal penalties works better to bring positive effect of corporate governance enforcement in emerging market context of India. In line with this argument, we suggest that it is not the adoption of CGR, but the accompanying stricter enforcement sanctions that could provide an improvement in prudent corporate governance practices. With more severe penalties, we expect the treated Indian firms to reduce their DPR as higher dividends are now less required to signal strong internal governance standards, which are now conveyed through the compliance of CGR. Therefore, we test our second hypothesis (H_2) to capture the effect of harsher enforcement sanctions.

H_2 : The introduction of the enforcement amendments in Section 23E reduces the DPR of treated firms more than those of control firms.

It is worth noting here that if harsher penalties are more effective than the introduction of CGR then we would expect the effect of CGR on DPR to be stronger in case of testing H_2 relative to H_1 .

4. Data and Variables

4.1. *Sample Data and Sources*

The sample used in this study is obtained from Prowess, a database maintained by the Center for Monitoring the Indian Economy. Prowess reports a comprehensive set of firm-level financial and market based variables for publicly-traded Indian companies.⁴ Our sample includes 4,000 firms with 21,932 firm-year observations over the period of 1997-2007. Our sample period is divided into two sub-periods. The first period ranges from 1997-2002 and the second covers 2002-2007. The first sub-period is intended to measure the introduction effect of Clause-49 in 2000 and the second sub-period captures the impact of the harsher sanctions (i.e. enforcement) of 2004. We exclude the financial and utilities firms as they follow different financial reporting standards, and both their DPR and their access to external capital markets are regulated (Renneboog and Trojanowski, 2011). We use the yearly panel with the fiscal year ending March 31st.

4.2. *Variables*

Table 1 summarizes the variables and how they are constructed. Our dependent variable of interest is the DPR which is the ratio of the sum of common and interim dividends to profit after tax, where the latter is used as a proxy for the firm's net income. Our two central independent variables of interest are the interaction of dummy variables for the two sub-sample periods. The first sub-sample includes an interaction term between the firm-specific indicator variable (TREAT) which takes the value of one if a firm is affected by the reforms and zero otherwise and a time-specific indicator variable (AFTER_CL49) that takes the value

⁴ The database has been used by a number of recent studies on Indian firms, including Lilienfeld-Toal et al. (2012), Vig (2013) and Gopalan et al. (2016).

of one from the year 2000 when Clause-49 was introduced otherwise zero. We define it as *DID_CL49* in our regressions. The second sub-sample incorporates an interaction term of TREAT and time-specific indicator variable (*AFTER_S23E*) that takes the value of one from the year 2005 after the year (2004) when harsher financial and criminal penalties were introduced and zero otherwise. We incorporate this interaction as *DID_S23E* in our empirical specifications. Our treated group is comprised of firms from Groups 1, 2 and 3, discussed in Section 2.1, including any firms that listed between 2000 and 2004. Our control group consists of Group 4 firms, which have paid-up equity capital of less than INR 30 million and thus are not subject to Clause 49 in 2000 or 2004.

Table 1 about here

Following previous empirical studies we also incorporate a number of controls. The first control is firm's size. Studies show that the size of a firm can play a key role in a firm's DPR (Denis and Osobov, 2008). Similarly, Fama and French (2001) report that firm size has a positive impact on dividends as larger firms tend to pay out more dividends. However, owing to the scale effect DPR can be negatively related to size. We measure firm size by taking the natural log of the market value of equity.

We control for a firms' capital structure (*Leverage*) as it has been shown that DPR are affected by financial constraints. The financial constraints argument suggests a positive relation between leverage and DPR. However, Brockman and Unlu (2009) argue that firms with higher leverage pay fewer dividends as creditors can pressure managers to reduce DPR and use the cash flows to service their debt instead. We measure leverage in our models as the ratio of net liabilities to total assets.

We also control for firm's profitability and growth opportunities. It is argued that firms with higher growth prospects should pay lower dividends as managers with higher

growth opportunities are expected to invest the proceeds of the firm into positive-NPV projects, indicating a negative relation between growth and DPR (Chay and Suh, 2009). However, to the extent that dividends send better signals to the market, especially in emerging markets with weaker information environment, the signalling hypothesis would suggest that higher DPR can be associated with higher valuation. For instance, Pinkowitz et al. (2006) finds that the relation between dividends and firm value is stronger in countries with weaker investor protection. Our measure of firm valuation is Tobin's Q and is computed as the ratio of the sum of total liabilities, book value of preferred stock and market value of equity to the book value of total assets. We use the book value, rather than the market value, because preference shares are traded very thinly in the Indian market. Similarly, profitable firms with higher return on assets (ROA) can afford to pay higher dividends with an expected positive sign between dividend payout and ROA. We measure ROA as the natural logarithm of the ratio of earnings before depreciation, interest and taxes to the book value of total assets. However, ROA can also proxy growth potential leading to the possibility of negative relation.

We control for volatility by computing yearly standard deviation as natural log of the daily stock returns. As riskier firms are likely to offer lower dividends, because of the perceived uncertainty of their profits and their overall performance in the market, the expected sign of this variable is negative (John and Knyazeva, 2006). We also control for the possible role of institutional investors: including foreign and domestic institutional investors. The literature argues that these investors actively monitor corporations worldwide. When their holdings are large they can pressurise management to pay higher DPR, thus limiting managers' scope to divert cash flows to themselves (Grinstein and Michaely, 2005; Ferreira and Matos, 2008). We include the percentage of company's stock held by foreign

institutional investors (*FII*). However, it is argued that their domestic counterparts often side with the managers and do not pressure them to pay higher dividends (Ferreira and Matos, 2008). As such, we also incorporate the percentage of company's stock held by domestic institutional investors (*DII*).

We also take account of the potential role of share-buybacks in explaining DPR. Share repurchases are increasingly viewed as viable substitutes of cash dividends, due to the tax advantage of capital gains in many jurisdictions. By opting for share repurchases, investors can delay the realization of any capital gains and subsequently the payment of taxes on these gains (Grullon and Michaely, 2002). We control for share-buybacks by creating a categorical variable taking value one if the firm has any repurchase activity and zero otherwise. Following Grinstein and Michaely (2005), we compute "Repurchase Activity" as the ratio of the nominal amount of repurchased equity to the book value of assets.

Finally, we control for firm fixed effects, year fixed effects and firm specific trends in DPR.⁵ To minimise the influence of obvious outliers we winsorize all firm fundamentals at the 5% and 95% level.

5. Empirical Results

We begin by exploring the summary figures followed by figurative examination of the yearly average DPR over the sample period. Following this we present the univariate and multivariable difference in differences (DID) estimations.

⁵ Controlling for firm specific trends besides firm and year fixed effects, is important because of the difference in our treated groups (large and medium sized firms) and smaller control firms there remains a possibility of difference in time trends in DPR driving our results, as oppose to our hypothesized claim of CGR.

5.1 Summary Statistics

Table 2 reports summary statistics of the dependent and the control variables. Panel A presents summary statistics for entire study period from 1997 to 2007 and panels B, C and D describe variables for pre-introduction period (1997-1999), post-Clause-49 introduction period (2000-2003) and post-23E sanction period (2005-2007) respectively. Panels A, B and C show that median firm pays zero dividends after 2000. There is a decreasing pattern in DPR over the period (0.20 in sub-period of 1997-1999, 0.16 in sub-period 2000-2003 and 0.14 in sub-period 2005-2007). This decrease provides some initial support for hypothesis 1 and could be indicative that following CGR in India DPR are decreasing. However, there are also significant changes in control variables over the study period. Size has increased from the pre-Clause-49 period to the post-23E period. Similarly, in comparison to 1997-1999, Tobin's Q shows slight decrease in the sub-period of 2000-2003 (0.89 to 0.86) but an increase in the post 23E period to 1.34. Volatility increases after 2005 (from 0.03 to 0.04). The institutional ownership (FII and DII) in our sample firms also shows an increase over time.⁶ In summary, we observe a general decline in dividend after the year 2000.

Table 2 about here

5.2 Trend of Yearly Average DPR

We also plot the demeaned yearly DPRs of both the treated and the control groups in Figure 2 and 3. Figure 2 and Figure 3 covers the three years pre and three years post the adoption of Clause-49 in the year 2000 and the imposition of stricter sanctions (section 23E) in the year 2004 respectively. Similar to the trend reported in summary statistic, we see a falling trend in

⁶ FII has the least number of observations, which is primarily because there is no data available on this variable prior to 2000.

DPR for both treated and control firms in Figure 1. One possible explanation for the fall after 1997 in India is the introduction of the Dividend Distribution Tax (DDT) in 1997, which taxes the issuing companies, instead of investors, on any paid-out cash dividends during a given year (PwC, 2017).

However, it is quite intuitive to argue that the effect of DDT alone would not last for long time. Even if it does have prolong effect we should expect the parallel trend to continue for both the treated and control firms in the absence of other factors that may possible change the course. The first such effect we identify is related to the introduction of Clause-49 in the year 2000. As we can see after 2000 the treated firms have in fact marginally increased their dividends compared to the dwindling trend of control firms. However, Figure 3 shows that in comparison to control firms that exhibit slight increase in dividend payments, the DPR of treated firms decreases drastically following the 2004 sanctions indicating that there is a possible impact of stricter sanctions (section 23E) on firms' dividend payout decisions.

Figure 2 about here

Figure 3 about here

5.3 Univariate DiD Analysis

Table 3 presents the univariate analysis for the two periods introduction of Clause-49 and amendment 23E. In Panel A, the DPR of both the treated and the control group decreases and are statistically significant. However, as shown in column 4, the difference in the differences of DPR between these two groups is statistically insignificant and economically immaterial. This suggests that the pre-2000 trend in both the treated and control groups continues post-2000 with and there is no difference between the groups. This indicates that the introduction

of Clause-49 may not have any effect on the treated firms' DPR relative to control firms. This could be due to the possibility that stakeholders did not show the expected positive response to the introduction of Clause-49 in the year 2000.

In Panel B of Table 3 which covers the 23E amendment we find there was virtually no change in the DPR of the control firms, but the treated firms' DPR fall by 1.6% (statistically significant at 1% level). The DiD of DPRs between the two groups is negative (-1.65%) and statistically significant. This lends support to our second hypothesis that the threat of harsher penalties for violating Clause-49 has led to a decline in the treated firms' DPR since outside investors now seem to exhibit greater trust on the possibility of compliance with the CGR.

Table 3 about here

5.4 Multivariate DiD Regressions

In testing the hypotheses we use two empirical specifications. The first, as shown in Equation 1, covers the CGR introduction period of 2000 i.e. sample ranges from 1997-2002:

$$DPR_{it} = \alpha + \beta.Treat_i.After_CL49_t + X_{it}.\delta + \gamma_i + \tau_t + g_it + e_{it} \quad (1)$$

where DPR is the dividend payout ratio for firm i in year t , $Treat_i$ a categorical variable that takes value one for firms affected by Clause-49 and $After_CL49_t$ is a dummy that takes the value one for year after 2000 and zero otherwise. The X_{it} are control variables as defined in Table 1 and discussed in section 4.2. The firm- and year-fixed effects are denoted by γ_i and τ_t respectively. g_it represents the firm-specific time trend in DPR, where g_i is the growth rate of DPR given by difference between $DPR_{i,t}$ and $DPR_{i,t-1}$ scaled by $DPR_{i,t-1}$ and $t =$

(1,2...N) is the linear trend variable for a period of 1997 to 2007, and e_{it} is a random error term.⁷

Our second model covers the enforcement period of 2004 i.e. from 2002-2007 as presented in Equation 2.

$$DPR_{it} = \alpha + \beta.Treat_i.After_S23E_t + X_{it}.\delta + \gamma_i + \tau_t + g_it + e_{it} \quad (2)$$

Here our main variable of interest is the interaction term, $Treat_i.After_S23E_t$, where $After_S23E_t$ is a dummy that takes value one for years after 2004 and zero otherwise. In addition to the above stated controls, we include now FII and DII in the set of controls i.e. X_{it} . The standard errors of all estimations throughout this study are clustered at firm levels. The results from the two empirical models (Equation 1 and 2) are reported in Table 4.

Table 4 about here

Columns [1] and [2] report regression outputs without and with control variables from equation (1) whereas columns [3] and [4] report those from equation (2) respectively. As we see in columns [1] and [2], the coefficients of Claus-49 in the year 2000 are not statistically significant, indicating that the introduction of the reform with only reputational penalties did not have any impact on DPR. This is inconsistent dividend substitution conjecture that CGR might replace DPR as governance mechanisms. Outside investors do not seem to be convinced by the adoption of Clause-49 that imposes greater transparency and disclosures. Consequently, firms do not respond by lowering their high levels of DPR as signal of their own internal corporate governance. The evidence shows that reputational penalty of delisting alone is not sufficient to substitute dividend as governance tools in an emerging market.

⁷ Since the data for FII and DII are only available after 2001, we are unable to control for include them in equation (1).

However, both the estimations that include the DID_S23E for the year 2004 yield significant ($p < 1\%$) negative coefficients as reported in columns ([3] and [4]) offering support for Hypothesis 2. In terms of economic magnitude, the results show a decrease of DPR of 3% to 5% in the treated firms relative to control firms. The results highlight the relevance of imposing additional personal penalties on corporate insiders for the violation of CGR provisions in emerging markets. Eventhough regulatory provisions in an emerging market imported from developed market are intended to substitute internal corporate governance signals, the outside investors and stakeholders are convinced about the improvement in governance enforcement only when CGR is accompanied by imposition of stringent personal penalties for violating Clause-49. The findings are consistent with the Becker's (1968) theory of punishment that suggests stronger punishment terms lead to higher compliance in weaker regulatory environment of emerging markets.

In terms of the effect of control variables, Size and ROA have been consistent in explaining negative effect on DPR in our sample firms for the study period. The positive coefficient of Tobin's Q indicates that firms with higher DPR are valued higher in the sample firms for the study period consistent with Pinkowitz (2006). The negative coefficient of ROA is indicative of the prevalent agency conflicts in Indian firms and the weak protection of minority shareholder (Young *et al.*, 2008). The effect of FII is positive and moderately significant suggesting higher foreign investors is associated with higher dividend pay-outs. Other control variables are insignificant.

5.5 Highly Comparable Groups

One concern of our multivariate DiD regressions as presented in Table 2 is a comparability issue as the treated firms in our model are larger in size in comparison to control firms. We address this issue by constructing comparable sub-samples of treated and control group firms. For creating comparable sub-samples of treated and control groups, we use the provision of Clause-49 which exogenously divides Indian firms into four different groups, identified in section 2.1.⁸ We present firm's characteristics (control variables) pre-section 23E sanctions (2002-2004) in Table 5 to identify comparable firms.

Table 5 about here

Table 5 (column 2, 3, 4 and 5) shows the treated firms (Group 1, 2, 3 and combined) are larger in comparison to their control counterparts. Specifically, within treated firms, Group 1 firms are larger firms, Group 2 are medium-sized companies whereas Group 3 firms are small-sized firms. Group 4 firms are small-sized firms control group not affected by Clause-49. From table 5, we note Groups 3 and 4 are mostly similar in size, ROA and Tobin's Q. Group 2 and Group 4 are comparable firms in their leverage. We therefore, construct two sub-samples of treated firms. The first is a size-matched sub-sample which comprises Group 3 as treated group and group 4 as control group. Second is leverage-matched treated sub-sample with Group 2 as treated group and Group 4 as control group. We conduct separate tests of our two hypotheses, where we restrict our analysis only to those two comparable groups of firms. The results are reported in Table 6.

⁸ Group 1 firms are large-cap companies listed as flag "A" category in Bombay Stock Exchange Ltd. (BSE). Group 2 firms are mid-cap companies that have paid-up capital greater than INR 100 million or net-worth greater than or equal to INR 250 million. Group 3 firms are low-cap firms that have paid-up capital between INR 100 million and 30 million. Groups 4 are control firms with paid-up capital less than 30 million. As defined in section 4, Groups 1-3 are subject to Clause-49 (treated group), whereas Group 4 is not (control group).

Table 6 about here

Columns 1 and 2 of Table 6 report regression results of size-matched sub-sample whereas columns 3 and 4 report leverage-matched sub-sample. Columns 1 and 3 correspond to the period of adoption of Clause-49 (i.e. 1997-2002), and columns 2 and 4 analyse the period of the imposition of stricter sanction (2002-2007). The results for Clause-49 remain unchanged from Table 4 i.e. Clause-49 fails to show an impact on DPR rejecting hypothesis 1. However, for Section 23E the results are even stronger than Table 4 showing a reduction of around 6% for size-matched firms and 5% for leverage matched firms in the treated firms' DPR. These results are stronger in both statistical significance and economic magnitude providing further support to hypothesis 2.

5.6 Further Robustness Checks

In this section, we discuss an array of robustness tests reducing confounders driving our results.

5.6.1. False Experiment Test

We suggest that the decline in the DPR of the treated firms can be attributed, at least in part, to Section 23E. However, this claim assumes that there are no important confounding events around 2004 that might have an impact on dividend policy. We therefore setup two false experiments, or placebo events, where we assume that Section 23E was enforced in October 2003 and 2005 respectively, instead of 2004. If there are any confounding events before or after 2004 that have strong effect on DPR, we would expect these false experiments to show significant effect like those reported for the year 2004.

The estimated effects, reported in Columns 1 and 2 of Table 7, show that both events are statistically insignificant. To limit any spill-over effects from the true experiment of 2004, we

run alternative checks by using only one year before and after the alternative enforcements of 2003 and 2005 – the DiD coefficients remain indistinguishable from zero. Essentially, the insignificance of the two false experiments reinforces the argument that it is Section 23E that has a negative impact on DPR.

Table 7 about here

5.6.2. For a narrower sub-period

Our initial results from Table 6 use of a three-year period around the enforcement of Section 23E, which might capture additional effects or events that occurred close to the enforcement year. Even though the false experiments provide some support that this is not the case, we use an alternative regression using a narrow period of 2 years before and after the enforcement of 2004 (i.e. 2003-2006). The DiD coefficient, as reported in Column 3 of Table 7, becomes stronger (the effect shows a 10% decline) both in significance and larger in magnitude in comparison to the 3 year-period.

5.6.3. First difference Regression

To further eliminate the role of unobservable factors explaining changes in DPR, in Column 4 of Table 6, following Dharmapala and Khanna (2013), we report the estimation using the first-differenced model on Equation 2, as expressed in Equation 3.

$$\Delta DPR_{it} = \beta \Delta Treat_i . After_S23E_t + \Delta X_{it} . \delta + \tau_t + g_i + e_{it} \quad (3)$$

where $\Delta DPR_{it} = DPR_{it} - DPR_{i,t-1}$; ΔX_{it} is a vector of control variables; g_i captures firm-specific trend in DPR; τ_t is the year effect and e_{it} is the error term. The DiD coefficient in Table 7 (Column 4) points to a statistically significant decrease of 9% in the treated firms'

DPRs following the enforcement of Section 23E, which further supports our second hypothesis. The results from the first-differenced model remain robust to conducting the regression only with the sub-sample of highly comparable treated and control groups (*results available from the authors on request*).

5.6.4. *Self-selection Issue.*

Self-selection problem in which firm can endogenously selected themselves to be affected or remain unaffected can undermine our empirical estimation. However, the applicability of Clause-49 is backward looking. For instance, if a firm has met the paid-up capital criteria at any point in the past, it is required to adhere to Clause-49 provision, even if it does not fall within the compliance bracket during the enforcement year. This helps reduce the possibility of self-selection of firms lowering their paid-up equity capital to avoid Clause-49 regulation. However, this provision does not rule out the possibility of firms increasing their paid-up equity capital to be affected by Clause-49. We therefore examine the stability of paid-up equity capital of the firms used in our empirical design and find the paid-up equity capital is these firms are highly stable for the study period consistent with Dharmapala and Khanna (2013). In an unreported table, we eliminate 32 firms that change their paid-up equity capital, re-run empirical estimations and book a finding consistent with our main results.

5.6.5. *Other concerns*

Finally, we briefly discuss some other alternative tests to check the robustness of our results. The outcomes are not reported but are available from the authors on request. One concern on our results is the entry of newly-listed firms in the treated group after 2004. These firms do not have matching observations in the pre-Section 23E period, which would inevitably increase the size of the treated group in the post-sanctions period (Dharmapala and

Khanna, 2013). Therefore, we remove any firms that are listed after 2004. Similarly, we run an additional regression, where all firms that were listed after 1999 are transferred to the control group. Our results largely remain unchanged to those reported in Table 4, with an insignificant DID_CL49 coefficient.

Another possible issue is that the DPR ratio includes both interim and final dividends, which have different signalling characteristics and most studies exclude the interim dividends when analysing dividend policy (La Porta et al., 2000; Adjaoud and Ben-Amar, 2010). Even though interim dividends are less frequently reported, they can signal to investors the potential payoff of the stock at the terminal date (Chen et al., 2014). Any cuts in the company's interim dividend are perceived by shareholders as more significant than the cuts in the final dividend, whereas the reverse is true for dividend increases (Aker, 1999). In other words, there is usually a peak in trading volatility after interim announcements of dividend cuts. When we include the interim dividends, we are therefore effectively adding an additional amount of public information to our model. Nevertheless, we find that the effect of the stricter penalties remains significant even after excluding the interim dividends.

Similarly, it can also be argued that our DPR ratio does not fully capture the typical dividend policy, since it does not include preferred dividends. Any omissions of preferred dividends are viewed by outside investors as signs of possible financial distress (Karels and Prakash, 1987). Our results for hypothesis 2 are robust to including preference dividends into the DPR.

Finally, firms in emerging markets can respond to the weak domestic regulatory environment by cross-listing into better-developed stock markets with stricter governance regulations. We identify a list of Indian firms that have cross-listed in Europe or the U.S.

through the issuance of depositary receipts ⁹ and our results are robust to the omission of these firms from the dataset.

Finally, concern with the enforceability of the CGR event 2004 in our setting, as is true with many other emerging economies, is that the sanctions could still struggle to translate from provision to practice with the existence of a weaker enforcement environment. This can undermine the credibility of our inference. However, the legal set-up for Clause-49 was such that enforcement under Section 23E would occur in the first instance by the SEBI with a potential appeal to the Securities Appellate Tribunal (a body formed to deal with securities laws issues and w addresses SEBI appeals), and followed by a final appeal to the Supreme Court. Reports suggest that the number (turnaround time) of settled cases on enforcement decisions has been increasing (decreasing) in the post enforcement periods on issues enforced by SEBI and the Securities Appellate Tribunal.¹⁰ Similarly, Balasubramanian et al. (2010) find that the majority of Clause-49 affected firms have complied with Clause-49 provisions in the post enforcement.¹¹ Taken together, 2004 CGR was introduced with a reasonably clear system for the prompt handling of cases for non-compliance, providing a credible improvement in the likelihood of enforcement.

5.7. Cross-sectional Heterogeneity and the Effect of Stricter Sanctions

In this sub-section, we provide further robustness checks by examining how Section 23E differentially affects firms depending on the differences in their DPR and leverage within the treated firms prior to 2004 enforcement.

⁹ The list of cross-listed Indian firms is obtained from www.adr.com (operated by JP Morgan Chase and Thomson Financial) and <https://www.adrbnymellon.com/directory/dr-directory> (operated by The Bank of New York Mellon).

¹⁰ Evidence can be found in reports such as Securities and Exchange of Board of India, Handbook of Statistics on the Indian Securities Market 2008, pp. 66-71 and SEBI, Annual Report 2007-08, pp. 103-114, 119-129.

¹¹ Balasubramanian et al. (2010) note that on an average there has been greater compliance with provisions of Clause-49, however, the compliance is far from universal.

5.7. DPR Heterogeneity and the Effect of CGR on DPR

To the extent that stricter CGR conveys credible improvement in corporate governance, DPR of high-dividend paying firms should decrease more after 2004. To explore how stricter corporate governance affects the heterogeneity of DPR we divide our sample firms into 3 terciles (33% percentiles) based on 3 year-average DPR before enforcement of stricter CGR (i.e. 2002-2004)¹² and examine how Section 23E differently affects the DPR of high dividend paying treated firms (the upper tercile firms) and low dividend paying (lower tercile firms) treated firms by using Equation 4:

$$DPR_{it} = \alpha + \beta.Treat_i . S23E_t . High - Paying_i + X_{it} . \delta + \gamma_i + \tau_t + g_{it} + e_{ijt} \quad (4)$$

where $Treat_i . S23E_t . High - Paying_i$ is the interaction term, $High - Paying_i$ is a categorical variable that takes value of one if the treated firm lies in upper tercile of DPR prior to section 23E enforcement and zero if treated firm is in lower tercile.

The result presented in column 1 of Table 8 shows that the DPR of high-dividend paying treated firms reduces 15% in comparison to low dividend paying treated firms following 2004. These findings highlight that the 2004 sanctions lowered DPR of high dividend-paying firms, who would otherwise have to pay higher dividends to signal better governance when market-based corporate governance is weaker (La Porta et al., 2000; Pinkowitz et al. 2006).

Table 8 about here

¹² We generate three-year average of dividend payout to generate heterogeneity among high dividend paying and low dividend paying firms following strategy adopted by Vig (2013).

5.8. *Leverage Heterogeneity and the Effect of CGR on DPR*

Dividend policies have an important influence on lenders' decision to invest in the firm, particularly in terms of their rights and whether they are ultimately repaid. Radner and Shepp (1996) argue that if managers pay out too much dividends (potentially as a signal), the cash reserve may drop below what is deemed as critical level by the lender and thus firms may find difficult to have access to the debt market. However, as noted, managers often choose to increase dividends as signal of better reputation among shareholders and to attract outside investors. This strategy is costly to lenders as it increases both the company's risk premium and bankruptcy costs (Fan and Sundaresan, 2000).

Conflicts between shareholders and lenders are especially acute in highly-leveraged firms, where the lenders have a high stake in the firm's cash flows. Thus, firms with higher leverage who pay higher dividends can find extremely difficult and/or expensive to access further debt. One solution to this problem is to have a corporate governance framework that may directly or indirectly reconcile any conflicts of interest between shareholders and creditors and ensures that the firm is not driven into default by managers who increase DPR (Claessens, 2006; Radner and Shepp, 1996). In other words, if CGR enhances the quality of their corporate governance and replaces DPR as a signal of firm quality then firms with higher leverage might reduce their dividends more than that of lower leverage. Therefore, we examine the cross-sectional heterogeneity of leverage and show how section 23E sanction affects high leverage firms in comparison to low leverage firms by using equation 5:

$$DPR_{it} = \alpha + \beta.Treat_i . S23E_t . \overline{Leverage_i} + X_{it} . \delta + \gamma_i + g_it + \tau_t + e_{it} \quad (5)$$

where $Treat_i . S23E_t . \overline{Leverage_i}$ is the interaction term, $\overline{Leverage_i}$ is 3 year average of leverage prior to section 23E-enforcement. Column 2 of Table 8 shows that the treated firms

with higher leverage have experienced a significant reduction in their DPR of 16% after 2004 in comparison to the treated counterparts with lower leverage. Essentially, consistent with the prediction of existing literature these results suggest that stricter sanctions benefit the lenders (Fan and Sundaresan, 2000; Radner and Shepp, 1996).¹³ Again, these results further support our key findings reported earlier.

6. Conclusion

The ‘substitution argument’ of dividend policy states mandatory regulatory provisions which improve the quality of corporate governance through reputational penalties imposed for non-compliance should substitute higher dividend payouts as governance tool. However, the literature also suggests that in the face of inadequate deterrence induced by regulatory interventions, particularly in emerging markets, CGR with reputational penalties may not yield the expected replacement effect in DPR. The substitution effect on dividend payouts may manifest itself only after the imposition of additional stricter personal liabilities of corporate insiders/managers in the criminal and financial penalties that would adequately induce higher compliance. We test the substitution effect of CGR on dividend policy by exploiting two corporate governance regulatory shocks of the same listing agreement in the Indian emerging market. The first regulatory shock in the year 2000 relates to the introduction of mandatory provisions of greater disclosure and board independence in their stock market listing agreement, popularly known as Clause-49. Since the penalty for non-compliance was delisting, this imposed reputational penalty when introduced. However, in

¹³ Section 23E indirectly complements the effect of the SARFAESI Act of 2002, whose purpose is to strengthen secured creditors’ rights (Vig, 2013). Claessens (2006) argues that an effective corporate governance framework protects creditors from expropriation and in this respect, Section 23E can be considered as a successful addition to Clause 49.

the year 2004 the regulators further introduced section 23E which imposed additional severe financial and criminal penalties for corporate insiders for violating Clause 49.

We use the difference-in-differences natural experimental method by exploiting both the regulatory shocks that exogenously generate treated and control firms. Using a sample of all listed firms spanning 1997-2007 our study shows that the introduction of Clause-49 that imposed significant reputational penalty in the year 2000 did not have any impact on the treated firms' dividend policies. However, the accompanying imposition of severe personal penalties on the corporate insiders in the form of economically large monetary fines and criminal charges in 2004 led to a material fall in the DPR of the treated firms. These findings are robust to a myriad of robustness checks, including ruling out the possibility of false experiment. The outcomes of our examination support dividend substitution hypothesis indicating that firms replace their dividend policies as governance tools but only after adequate penalties for deterring non-compliance are imposed. Our study supports the view of expanding personal accountability to corporate decision makers to promote improve the effectiveness of CGR enforcements.

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Table 1. List of the Variables Used in the Analyses

This table shows the construction of variables. Explanations are provided in the description of variables in the text.

S. No	Variables	Source
	<i>Dependent Variable</i>	
1	DPR = Cash Dividends/Profit after Tax	CMIE
	<i>Independent Variables</i>	
1	DID_CL49 = TREAT*AFTER_CL49	Own Calculation
2	DID_S23E = TREAT*AFTER_S23E	Own Calculation
3	DIDID_High_paying= TREAT*AFTER_S23E*High – Paying _i	Own Calculation
4	DIDID_Leverage= TREAT*AFTER_S23E*Leverage _i	Own Calculation
	<i>Control Variables</i>	
1	Size = ln (market value of equity)	Derived from CMIE
2	Leverage = $\frac{\text{Net Total Liabilities}}{\text{Total Assets}}$	Derived from CMIE
3	ROA = $\ln(1 + \frac{\text{PBDITA}}{\text{Total Assets}})$	Derived from CMIE
4	Tobin's Q = $\frac{\text{Total debt} + \text{BV(Preferred Stock)} + \text{MV(Equity)}}{\text{BV(Total Assets)}}$	Derived from CMIE
5	VOLATILITY = 365-day average of standard deviation of natural log of daily stock returns	Derived from CMIE
6	FII = % Shares held by: foreign corporate bodies, foreign institutions, qualified foreign institutions, foreign venture capital funds	Derived from CMIE
7	DII = % Shares held by: Indian corporate bodies; financial institutions and banks; mutual funds; insurance	Derived from CMIE
8	Buy-back=1 if firm has any repurchase activity and 0 otherwise	Derived from CMIE
9	Firm specific trends in DPR = g _i * t	Derived from CMIE
	<i>Ancillary variables</i>	
1	TREAT = 1 if a firm is affected by Clause-49 and 0 otherwise	Derived from CMIE
2	AFTER_CL49 = 1 if year>2000 and 0 otherwise	Derived from CMIE
3	AFTER_S23E = 1 if year>2004 and 0 otherwise	Derived from CMIE
4	Net Total Liabilities = Total Liabilities – Total Capital – Reserves and Funds – Share application money and suspense account	Derived from CMIE
5	Book Value (Equity) = Total Assets – Net Total Liabilities	Derived from CMIE
6	Market Value (Equity) = (365-days average of daily stock price) × (365-days average of number of shares outstanding)	Derived from CMIE
7	g _i = firm-specific growth in DPR = $\frac{DPR_{i,t} - DPR_{i,t-1}}{DPR_{i,t-1}}$	Derived from CMIE
8	High – Paying _i = 1 if firm lies in upper tercile of DPR prior to section 23E enforcement (2002-2004) and zero if treated firm is in lower tercile.	Own Calculation
9	Leverage _i = 3 year average of Leverage	Derived from CMIE

Table 2: Summary Statistics

This table shows the summary statistics (number of observation, mean, median, standard deviation, minimum and maximum values) of all the variables used in the analysis. Panel A reports summary statistics of variables for entire study period (1997-2007) whereas panels B to D report those for sub-periods including pre-Clause-49 period (1997-1999), post-Clause-49 period (2000-2004) and post 23E sanction period (2005-2007) respectively. DPR is calculated by dividing the sum of the interim and the final dividends by the profit after tax. Size is calculated taking the natural logarithm of the market value of equity. Leverage is the ratio of net liabilities to total assets. ROA is computed as a natural logarithm of the ratio of earnings before depreciation, interest and taxes to the book value of total assets. Tobin's Q is the ratio of the sum of total liabilities, book value of preferred stock and market value of equity to the book value of total assets. Volatility is the yearly standard deviation of the natural log of daily stock returns. FII is the percentage of firm's stock held by foreign institutional investors (FII). DII incorporates the percentage of firm's stock held by domestic institutional investors (DII). Buy-back dummy takes value one if the firm has any share-repurchase activity in a given year and zero otherwise. Calculations of the variables are described in Table 1. Source: CMIE.

	Count	Mean	Median	St. dev	Min	Max
Panel A (1997-2007)						
DPR	21932	0.17	0.00	0.25	0.00	1.27
Size	16384	5.36	5.12	1.99	2.08	9.17
Leverage	21932	0.54	0.55	0.29	0.04	2.04
ROA	21932	0.13	0.12	0.06	0.00	0.24
Tobin's Q	21932	1.01	0.82	0.71	0.12	3.33
Volatility	16068	0.07	0.05	0.06	0.02	0.29
FII	4146	0.06	0.01	0.09	0.00	0.34
DII	9131	0.12	0.05	0.16	0.00	0.53
Buy-back dummy	21932	0.01	0.00	0.09	0.00	1.00
Panel B (1997-1999)						
DPR	7875	0.20	0.10	0.27	0.00	1.27
Size	6293	4.85	4.52	1.86	2.08	9.17
Leverage	7875	0.52	0.54	0.23	0.04	2.04
ROA	7875	0.13	0.12	0.06	0.00	0.24
Tobin's Q	7875	0.89	0.76	0.61	0.12	3.33
Volatility	6160	0.10	0.06	0.08	0.02	0.29
FII	-	-	-	-	-	-
DII	-	-	-	-	-	-
Buy-back dummy	7875	0.00	0.00	0.04	0.00	1.00
Panel C (2000-2004)						
DPR	7780	0.16	0.00	0.25	0.00	1.27
Size	5400	5.16	4.87	1.96	2.08	9.17
Leverage	7780	0.53	0.55	0.29	0.04	2.04
ROA	7780	0.12	0.12	0.06	0.00	0.24
Tobin's Q	7780	0.86	0.76	0.58	0.12	3.33
Volatility	1725	0.03	0.00	0.06	0.00	0.34
FII	4468	0.07	0.03	0.09	0.00	0.53
DII	7780	0.01	0.00	0.11	0.00	1.00
Buy-back dummy	6277	0.01	0	0.10	0	1
Panel D (2005-2007)						
DPR	6277	0.14	0.00	0.21	0.00	1.27
Size	4691	6.28	6.22	1.90	2.08	9.17
Leverage	6277	0.59	0.58	0.36	0.04	2.04
ROA	6277	0.13	0.12	0.07	0.00	0.24
Tobin's Q	6277	1.34	1.10	0.85	0.12	3.33
Volatility	4650	0.04	0.04	0.02	0.02	0.29
FII	2421	0.08	0.03	0.11	0.00	0.34
DII	4663	0.17	0.09	0.19	0.00	0.53
Buy-back dummy	6277	0.01	0.00	0.10	0.00	1.00

Table 3: Univariate DiD

This table presents the difference of the DPRs of control and treated firms before and after the adoption of Clause-49 in 2000 in panel A and imposition of Section 23E in 2004 in panel B. DPR is calculated by dividing the sum of the interim and the final dividends by the profit after tax. The control group is comprised of firms unaffected Clause-49 reform, whereas the treated group is comprised of firms affected by the reform. DiD captures the difference in the differences, which is computed by subtracting the difference in the DPR of the control group from the difference in the DPR of the treated group. Source: CMIE

<i>Panel A: 1997-2002</i>	Pre-Clause-49	Post-Clause-49	Difference	t-stat	No. of Obs.
DPR (Control)	0.1761	0.1229	-0.0532	-4.90	1517
DPR (Treated)	0.1901	0.1442	-0.0459	-8.53	10056
Difference in differences (DiD)			-0.0073	1.49	
<i>Panel B: 2002-2007</i>	Pre-23E Sanction	Post-23E Sanction	Difference	t-stat	No. of Obs.
DPR (Control)	0.1018	0.1021	0.0004	0.05	2096
DPR (Treated)	0.1490	0.1329	-0.0161	-4.17	10105
Difference in differences (DiD)			-0.0165	-4.79	

Table 4. Difference-in-Difference Analysis – Basic Results

This table shows the results of four fixed-effects panel regressions as shown in equations 1 and 2 in the text. The dependent variable is the Dividend Payout Ratio. The independent variable of interest in Columns 1 and 2 is DID_CL49 – an interaction term between an indicator variable, $Treat_i$, which takes value of one for firms affected by Clause-49 and zero otherwise and an event indicator variable, $After_CL49_t$, which takes value one for years after Clause-49 was applicable (2001-2003) and zero otherwise. The main independent variable of Columns 3 and 4 is DID_S23E – an interaction term between an indicator variable, $Treat_i$, which takes value of one for firms affected by Clause-49 and zero otherwise and an event indicator variable, $After_S23E_t$, which takes value one for years after Section 25E was applicable (2005-2007) and zero otherwise. Firm controls include Size, Leverage, ROA, Tobin's Q, Volatility, FII, DII, Buy-back dummy and firm- specific trends. Variables are as defined in the notes to table 2 and winsorized at 5% and 95%. Year FE is Year Fixed Effects and Firm FE is Firm Fixed Effects. t-statistics are reported in parentheses. Standard errors are clustered at firm level and the levels of significance are indicated as follows: * at 10%; ** at 5% and *** at 1%. Data source: CMIE

	Dependent Variable: Dividend Payout Ratio (DPR)			
	[1]	[2]	[3]	[4]
DID_CL49 [$Treat_i . After_CL49_t$]	0.02 (0.64)	0.03 (1.27)		
DID_S23 [$Treat_i . After_S23E_t$]			-0.03*** (-3.06)	-0.05*** (-4.17)
ROA		-0.015*** (-5.29)		-0.014*** (-5.30)
Volatility		-0.001 (-0.87)		-0.001 (-0.15)
Size		-0.07*** (-6.69)		-0.05*** (-3.42)
Leverage		0.01 (0.17)		0.09 (1.63)
Tobin's Q		0.01** (2.05)		0.01* (1.83)
Buy-back dummy		0.01 (0.24)		0.03 (0.75)
FII				0.12* (1.71)
DII				-0.03 (-0.82)
Year FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Firm specific trend	YES	YES	YES	YES
Adj. R ² (within)	0.02	0.24	0.10	0.24
No of Firms	1201	1201	805	805
No of observation	7036	7036	3089	3089

Table 5. Firm-characteristics of Treated and Control Groups

This table shows mean and standard deviation of control variables of the sample firms classified into four different groups based on applicability of Clause-49 for pre-enforcement period (2002-2004). Variables are defined in the notes to Table 2. Group 1 firms are large-cap companies listed as flag "A" category in Bombay Stock Exchange Ltd. (BSE). Group 2 firms are mid-cap companies that have paid-up capital greater than INR 100 million or net-worth greater than or equal to INR 250 million. Group 3 firms are low-cap firms that have paid-up capital between INR 100 million and 30 million. Groups 4 are control firms with paid-up capital less than 30 million. As defined in section 4, Groups 1-3 are subject to Clause-49 (treated group), whereas Group 4 is not (control group). Source: CMIE

Category		Treated Group (1, 2, 3 and combined)			Control Group (4)
Clause-49 Group	Group1	Group 2 (Leverage-Matched Treated Group)	Group 3 (Size Matched Treated Group)	Groups 1, 2 & 3 combined	Group 4
Size	8.38 (1.03)	6.01 (1.50)	3.68 (1.05)	5.38 (1.98)	3.68 (1.18)
Leverage	0.54 (0.19)	0.58 (0.26)	0.53 (0.31)	0.55 (0.28)	0.58 (0.37)
Tobin's Q	1.56 (0.81)	0.98 (0.53)	0.79 (0.52)	0.91 (0.61)	0.77 (0.51)
ROA	0.15 (0.05)	0.13 (0.05)	0.11 (0.06)	0.13 (0.06)	0.11 (0.06)
Volatility	0.03 (0.01)	0.05 (0.03)	0.09 (0.03)	0.07 (0.04)	0.08 (0.04)
FII	0.07 (0.08)	0.02 (0.03)	0.03 (0.04)	0.04 (0.06)	0.04 (0.04)
DII	0.13 (0.10)	0.07 (0.08)	0.04 (0.06)	0.06 (0.09)	0.06 (0.09)
No of firms	126	828	870	1824	438

Table 6: DiD Panel Regression – Alternative Treated and Control Groups

This table shows the results of four fixed-effects panel regressions as shown in equations 1 and 2 in the text. The dependent variable is the Dividend Payout Ratio. Columns 1 and 2 presents results for size-matched sub-sample and columns 3 and 4 reports results for leverage-matched sub-sample. DID_CL49 is an interaction term between an indicator variable, $Treat_i$ which takes value of one for firms affected by Clause-49 and zero otherwise (excluding newly-listed firms after 2004) and an event indicator variable, $After_CL49_t$ which takes value one for years after Clause-49 was applicable (2001-2003) and zero otherwise. DID_S23E is an interaction term between an indicator variable, $Treat_i$ which takes value of one for firms affected by Clause-49 and zero otherwise (excluding newly-listed firms after 2004) and an event indicator variable, $After_S23E_t$ which takes value one for years after Section 25E was applicable (2005-2007) and zero otherwise. Firm controls include Size, Leverage, ROA, Tobin's Q, Volatility, FII, DII, Buy-back dummy and firm- specific trends. Variables are as defined in the notes to table 2 and winsorized at 5% and 95%. Year FE is Year Fixed Effects and Firm FE is Firm Fixed Effects. t-statistics are reported in parentheses. Standard errors are clustered at firm level and the levels of significance are indicated as follows: * at 10%; ** at 5% and *** at 1%. Data source: CMIE

Treated Control	Size-Matched Group 3 and Group 4 Firms		Leverage Matched-DiD Group 2 and Group 4 Firms	
	[1]	[2]	[3]	[4]
DID_CL49 [$Treat_i . After_CL49_t$]	-0.03 (-0.81)		0.02 (0.55)	
DID_S23E [$Treat_i . After_S23E_t$]		-0.06*** (-4.22)		-0.05*** (-3.07)
Size	-0.09 (-1.19)	-0.01 (-0.25)	-0.11*** (-5.41)	-0.07** (-3.53)
Leverage	0.04 (0.21)	-0.07 (-0.60)	0.09 (1.10)	0.07 (0.95)
ROA	-0.01* (-1.79)	-0.02* (-1.74)	-0.02* (-1.95)	-0.01* (-1.87)
Tobin's Q	0.04 (1.04)	0.03 (1.28)	0.02 (1.49)	0.02* (2.40)
Volatility	-0.27 (-0.97)	0.81 (0.22)	-0.05 (-0.24)	-0.46 (-0.51)
FII		-0.30 (-1.21)		0.10 (1.43)
DII		0.04 (0.46)		-0.01 (-0.31)
Buy-back dummy	-0.15 (-1.11)	-0.03 (-0.33)	-0.01 (-0.20)	0.00 (0.12)
Firm Specific Trend	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Adj. R ² (within)	0.27	0.28	0.26	0.26
No of firms	268	172	775	387
No of observation	878	474	2886	1432

Table 7: Difference-in-Differences Analysis – Robustness Checks

This table shows the results of four fixed-effects panel regressions. The dependent variable is the Dividend Payout Ratio and all variables are as reported in Table 1. Columns 1 and 2 report DiD regression results for two false experiments with one year before and one year after the true event year, 2004. Column 3 reports DiD regression for alternative (narrow) period and column 4 presents results from first difference regression as shown in equation 3. DID_B is an interaction term between $Treat_i$ which takes value of one for firms affected by Clause-49 and zero otherwise, and categorical variable of false-event, $After_FE_{2003}$, which takes value of one for year 2004-2005, and zero otherwise. DID_A is an interaction term between $Treat_i$ and a dummy variable with false event year, $After_FE_{2005}$ which takes value of one for years 2006-2007. DID_S23E is an interaction term between $Treat_i$ and the true event dummy, $After_S23E_t$ which takes value one for years after Section 23E was applicable (2005-2006) and zero for years 2003 and 2004. The coefficients in the fourth column represent the results of first difference regression. Firm controls include Size, Leverage, ROA, Tobin's Q, Volatility, FII, DII, Buy-back dummy and firm- specific trends. Variables are as defined in the notes to table 2 and winsorized at 5% and 95%. Year FE is Year Fixed Effects and Firm FE is Firm Fixed Effects. t-statistics are reported in parentheses. Standard errors are clustered at firm level and the levels of significance are indicated as follows: * at 10%; ** at 5% and *** at 1%. Data source: CMIE.

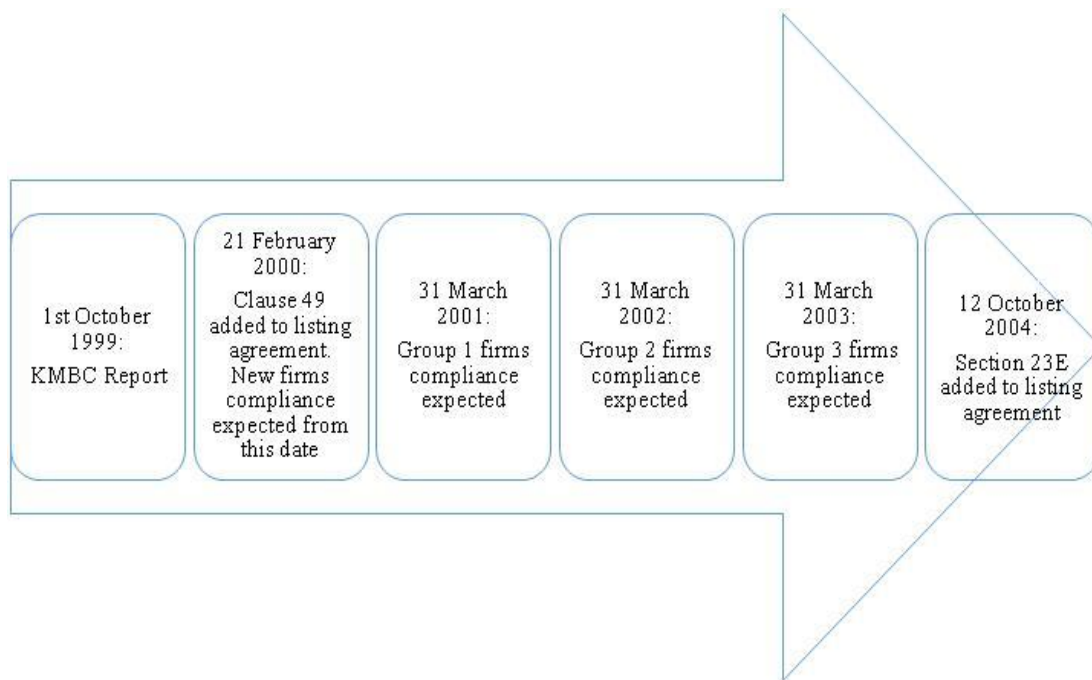
	False Experiment for 2003	False Experiment for 2005	Alternative Period 2003-2006	First Differences for 2002-2007
	[1]	[2]	[3]	[4]
DID_B [$Treat_i . After_FE_{2003}$]	-0.02 (-0.45)			
DID_A [$Treat_i . After_FE_{2005}$]		0.05 (1.34)		
DID_S23E [$Treat_i . After_S23E_t$]			-0.10*** (-3.55)	-0.09*** (-5.33)
Size	-0.05 (-2.05)	-0.03*** (-3.54)	-0.03*** (-3.25)	-0.03*** (-3.01)
Leverage	0.12 (1.36)	0.11** (2.08)	0.15* (1.95)	0.13** (2.26)
ROA	-0.009 (-1.56)	-0.012*** (-5.89)	-0.009** (-4.40)	-0.007*** (-4.28)
Tobin's Q	0.02 (1.10)	0.02 (1.99)	0.03* (2.90)	0.02*** (1.63)
Volatility	-0.001 (-0.09)	-0.002 (-0.19)	0.003 (0.75)	0.009 (1.64)
FII	0.29 (1.56)	0.06 (1.05)	0.10 (1.22)	-0.05 (-0.73)
DII	0.09 (0.49)	-0.04 (-1.25)	-0.02 (-0.68)	-0.02 (-0.73)
Buy-back dummy	0.00 (0.04)	-0.01 (-0.31)	-0.02 (-0.88)	-0.01 (-0.03)
Firm-specific trend	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	NO
Adj. R ² (within)	0.39	0.37	0.41	0.42
No of firms	512	755	695	583
No of observation	1327	1906	1638	1814

Table 8: Cross-sectional heterogeneity and DPR

This table shows the results of DIDID regression. Column 1 reports the heterogeneity effect of 2004 sanctions on high and low dividend paying firms whereas column 2 presents the effect of 2004 sanctions on leverage heterogeneity. DIDID-High paying is a triple interaction term $Treat_i \cdot After_S23E_t \cdot High_paying_i$ where $Treat_i$ takes value one for firms affected by Clause-49 and zero otherwise, $After_S23E_t$ is a dummy variable taking value one for period 2005 to 2007 and zero otherwise and $High_paying_i$ is a categorical variable that takes value one for high dividend paying firms (upper tercile) and zero for low dividend-paying firms (lower tercile) based on 3 year average of DPR for a period of 2002 to 2004. $Leverage_i$ is the 3-year average of firm's leverage prior to section 23E enforcement. Firm controls include Size, Leverage, ROA, Tobin's Q, Volatility, FII, DII, Buy-back dummy and firm- specific trends. Variables are as defined in table 1 and winsorized at 5% and 95%. Year FE is Year Fixed Effects and Firm FE is Firm Fixed Effects. t-statistics are presented in the parentheses. Standard errors are clustered at firm level and the levels of significance are indicated as follows: * at 10%; ** at 5% and *** at 1%. Data source: CMIE. Study period: 2002-2007.

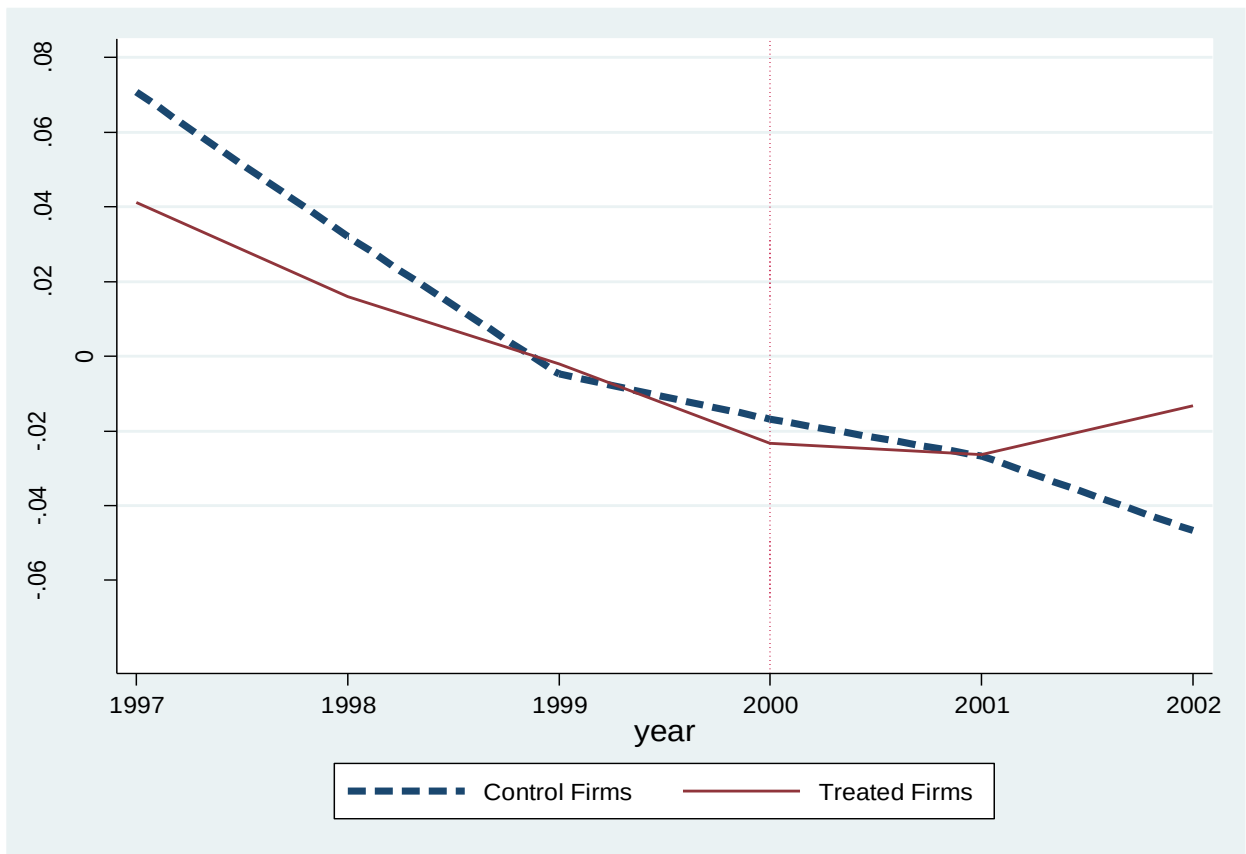
	Payout Heterogeneity	Leverage Heterogeneity
	[1]	[2]
DIDID-High paying [$Treat_i \cdot After_S23E_t \cdot High_paying_i$]	-0.15*** (-6.67)	
DIDID-Leverage [$Treat_i \cdot After_S23E_t \cdot Leverage_i$]		-0.16*** (-2.89)
Size	-0.01 (-1.20)	-0.03** (-2.40)
Leverage	0.06 (1.22)	0.07 (1.10)
ROA	-0.018*** (-6.63)	-0.012*** (-5.82)
Tobin's Q	0.02* (1.95)	0.02 (1.62)
Volatility	-0.012** (-2.30)	-0.001 (-0.20)
FII	0.002** (2.56)	0.001 (1.02)
DII	-0.01 (-0.28)	-0.03 (-1.04)
Buy-back dummy	-0.01 (0.57)	-0.01 (-0.49)
Year FE	YES	YES
Firm FE	YES	YES
Firm specific trend	YES	YES
Adj. R ² (within)	0.38	0.38
No of Firms	679	795
No of observation	2684	3069

Figure 1. Implementation Schedule of Clause 49.



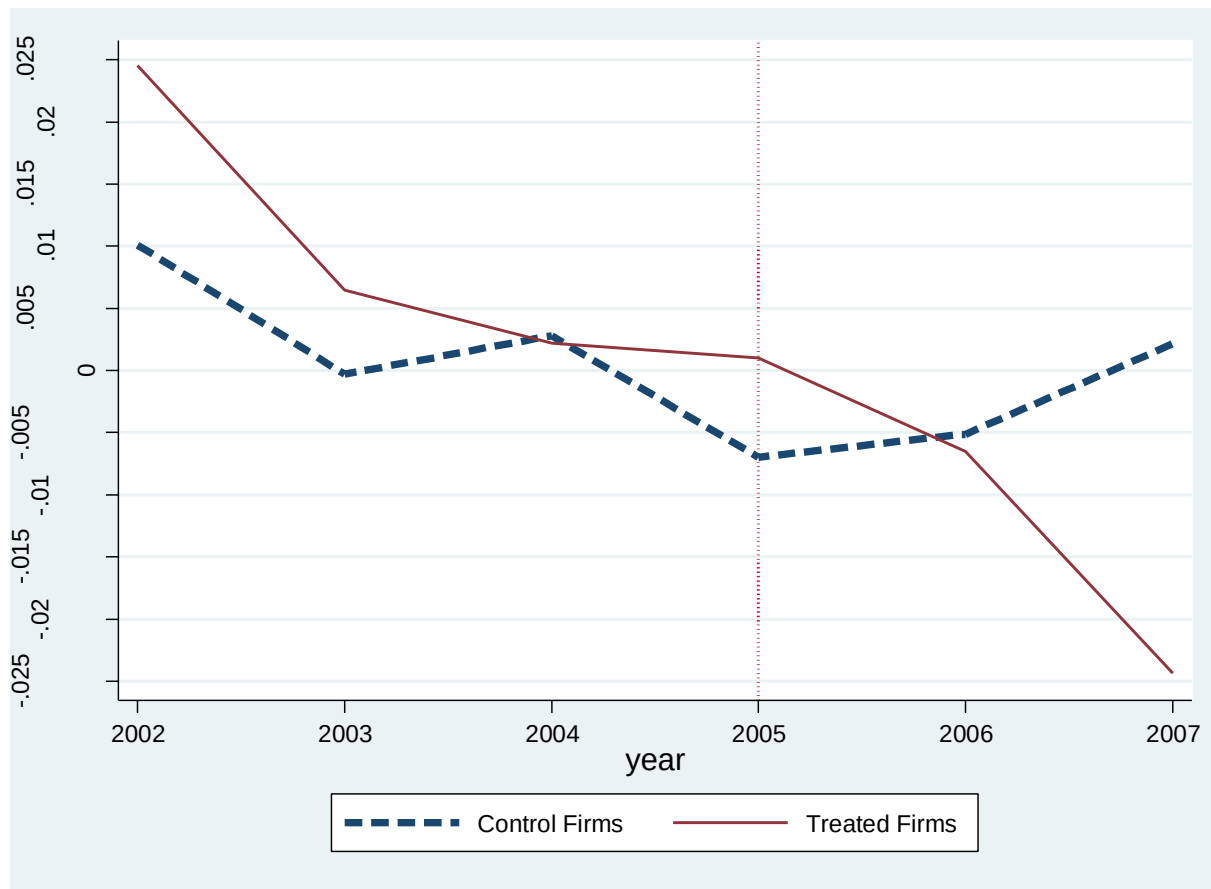
Note: Here a timeline of the Clause-49 and Section 23E enforcement is presented.

Figure 2: Dividend Payout Ratios and introduction of Clause-49



Note: The graph shows the demeaned value of annual Dividend Payout Ratios of treated and control groups of Clause-49 regulation for period of 1997-2002. The Dividend Price Ratio (DPR) is as the ratio of total dividend paid to total profit after tax as defined in Table 1. Treated Group firms are defined as those with paid-up equity capital of more than INR 30 million or with net worth of more than 250million as of or before 2004, or those that were listed after April 2000. All other firms are classified as Control Group firms. The vertical bar represents the year of enactment of Clause-49.

Figure 3: Dividend Payout Ratios and adoption of Section 23E



Note: The graph shows the demeaned value of annual Dividend Payout Ratios of treated and control groups of Clause-49 regulation for period of 2002-2007. The Dividend Price Ratio (DPR) is as the ratio of total dividend paid to total profit after tax as defined in Table 1. Treated Group firms are defined as those with paid-up equity capital of more than INR 30 million or with net worth of more than 250million as of or before 2004, or those that were listed after April 2000. All other firms are classified as Control Group firms. The vertical bar represents the year of adoption of Section 23E.